Assignment 1: Opinion Network Formation

Team Name: Complex Network of Dynamic People

September 18, 2026

GitHub Link to Code

`https://github.com/ShreeLite/DPCN_A1`

1 Dataset Documentation

The data cleaning and encoding pipeline described below was implemented as part of the network construction workstream (see `build_network.py`); this section documents and elaborates on that process.

The raw survey collected responses from 96 students across 60 statements on a 5-point Likert scale (Strongly Disagree, Disagree, Neutral, Agree, Strongly Agree), evenly split into four topics of 15 items each: Technology (prefix `T`), Education (`E`), Ethics/Society (`S`), and Environment (`V`). Before any analysis, the raw responses were cleaned in two stages. First, the string "No Comments" was treated as a genuinely missing response rather than a neutral ("Neutral") opinion, since a respondent skipping a question and a respondent actively choosing the midpoint of the scale are not the same thing. Second, respondents who left more than 50% of the 60 items blank were removed entirely (9 of the original 96 respondents), since a profile that is mostly missing cannot be meaningfully placed in an opinion network – this left 87 usable respondents. Any remaining sparse gaps (a respondent missing a handful, but not most, of the items) were filled with that item's class-average response, a conservative choice that avoids manufacturing an artificially strong opinion where none was expressed. Finally, the ordinal Likert scale was encoded numerically as Strongly Disagree $= -2$, Disagree $= -1$, Neutral $= 0$, Agree $= +1$, Strongly Agree $= +2$, producing the 87×60 numeric matrix (`outputs/encoded_responses.csv`) that both the person-opinion and item-opinion networks in this report are built from.

Descriptive Statistics and Data Exploration

To understand what the cleaned data actually looks like before any network is built, Table 1 summarizes the encoded response distribution for each of the four topics (pooling all 87 respondents $\times$ 15 items per topic).

Every topic has a positive mean, i.e. the class leans toward agreement on nearly every statement, but the four topics are not equally consensual. Environment and Ethics/Society have both the highest means *and* the lowest standard deviations – around 88% of all responses to these items are Agree or Strongly Agree, and almost nobody disagrees. Education has the lowest mean and by far the highest spread (std. 1.15): the most contested individual items in the entire survey are all Education items concerning assessment and attendance policy (e.g. "class attendance should be

Topic	Mean	Std. Dev.	Median	% Agree+	% Disagree-
Environment	1.39	0.75	2	89.7%	2.3%
Ethics/Society	1.28	0.83	1	86.1%	3.7%
Technology	1.00	0.99	1	74.7%	8.2%
Education	0.91	1.15	1	70.4%	13.7%

Table 1: Per-topic descriptive statistics of the encoded response matrix (-2 = Strongly Disagree, $+2$ = Strongly Agree). "% Agree+" and "% Disagree−" give the share of all (respondent, item) values in that topic that were Agree/Strongly Agree or Disagree/Strongly Disagree respectively.

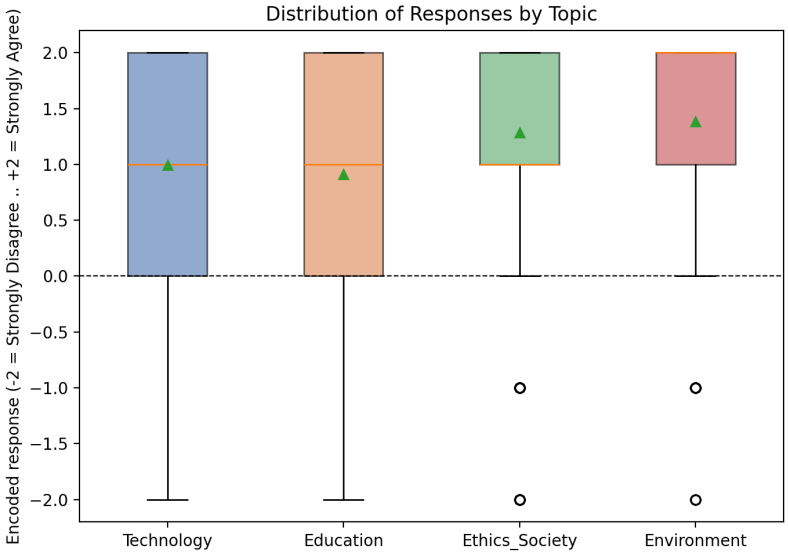


Figure 1: Distribution of encoded responses by topic. All four topics skew positive; Education has the widest spread and the only topic whose interquartile range dips to Neutral.

compulsory", "online learning can effectively complement classroom teaching"), where personal experience plausibly differs far more between students than it does for abstract ethics or environment statements. The most consensual items, by contrast, are broad normative statements (e.g. "continuous learning and skill development are essential", "companies should be held accountable for the environmental impacts of their activities") that a Likert survey rarely gets genuine disagreement on.

A note on self-report survey bias. This skew toward agreement is exactly what we would expect if several well-known biases in self-report surveys are present here. **Social desirability bias** means respondents tend to agree with statements framed as normatively "good" positions – sustainability, ethics, personal responsibility – regardless of their private attitude, particularly in a class survey where responses are not fully anonymous to the respondent's peer group. **Acquiescence bias** (a general tendency to agree with whatever is presented) would independently push every topic's mean upward, and is consistent with the fact that even the least-agreed-with topic (Education) still has a positive mean. **Self-selection** in who chose to complete the survey at all may also matter: 9 of 96 respondents were dropped for leaving over half the survey blank, and the remaining 87 who engaged with all 60 statements may be more opinionated or more agreeable

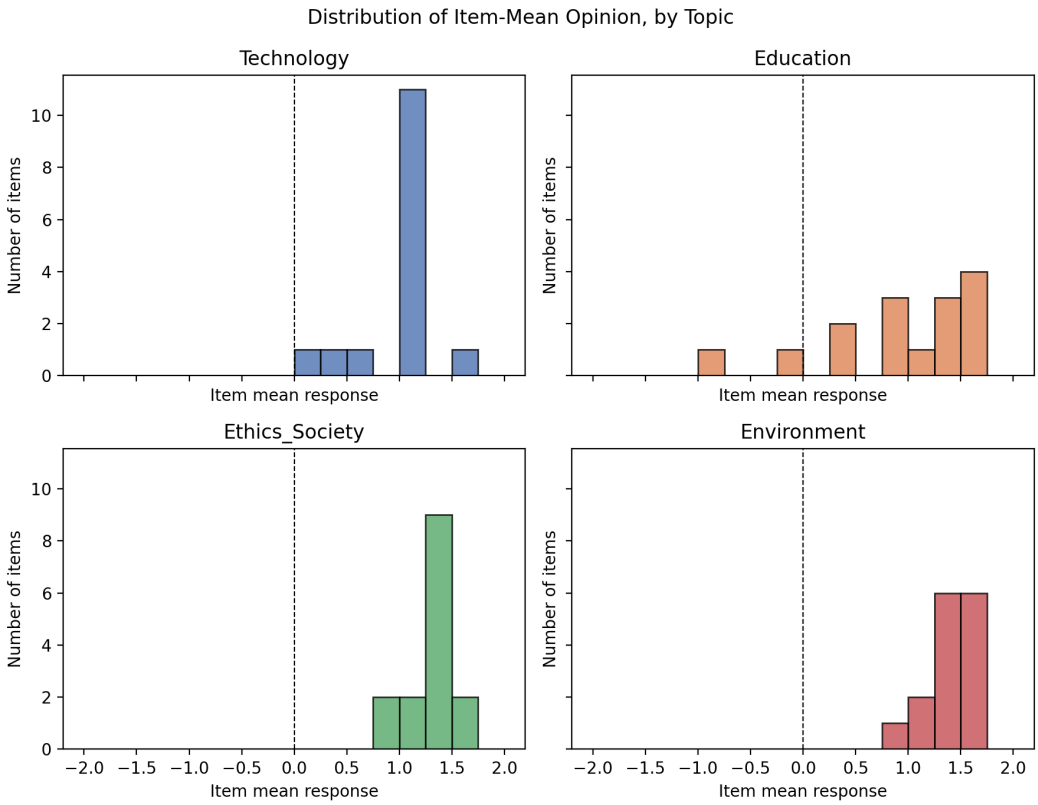


Figure 2: Distribution of per-item mean opinion (60 items total), faceted by topic. Technology and Environment items are tightly concentrated around a single positive mean; Education items are the most dispersed, including the only two topic-level items with a negative mean.

than a true random sample of the class. Because of this skew, raw agreement levels should be read with caution, and the *relative* structure captured by correlation-based similarity – used throughout both the person-opinion network (Section 2) and the item-opinion network (Section 2.5) – is a more robust signal of real opinion differences than absolute response levels are.

2 Pipeline Followed

This section describes how the raw survey responses were transformed into an opinion network.

2.1 Data Cleaning and Encoding

Respondents who left more than 50% of the survey unanswered were removed (9 of 96 respondents), leaving 87 usable respondents. The string "No Comments" was treated as a missing value rather than a neutral opinion. Likert responses were mapped to an ordinal numeric scale from -2 (Strongly Disagree) to $+2$ (Strongly Agree), and any remaining sparse gaps were filled with that item's class-average response.

2.2 Similarity Measure

Pairwise opinion similarity between respondents was computed using the **Pearson correlation coefficient** across their answer vectors, rather than cosine similarity or Euclidean distance. Correlation was chosen because it captures whether two respondents' opinions rise and fall *together* across statements, and is invariant to a respondent's individual tendency to answer more or less extremely overall – a scale bias rather than a genuine opinion difference.

2.3 Graph Construction

The dense 87×87 similarity matrix was sparsified using **mutual k-Nearest-Neighbours** with $k = 8$: an edge between respondents i and j was retained only if each was among the other's top-8 most similar peers. This produces a graph reflecting genuinely reciprocal opinion closeness, avoids an uninformative fully-connected graph, and sidesteps the need to hand-tune a similarity threshold.

In addition to the overall network (built from all 60 items), four topic-specific sub-networks were constructed using the same procedure restricted to each topic's 15 items, to allow cross-topic comparison of clustering structure.

2.4 Community Detection

Louvain modularity maximization was applied to the overall network to identify opinion communities. The same procedure was applied independently to each topic sub-network for comparison.

[TODO: PRATHISH (Statistical Validation): briefly describe your null-model validation methodology here (1 short paragraph) – e.g. "To assess whether the observed community structure is statistically meaningful, modularity was compared against a null distribution generated from N random graphs matched on node count, edge count, and degree sequence..."]

2.5 Item-Opinion Network Construction

In addition to the person-opinion network above, a second network was built with the opposite orientation: nodes are the 60 survey *items* rather than the 87 respondents, and an edge reflects whether two items were answered similarly across respondents – i.e., whether people who agreed strongly with one statement also tended to agree strongly with the other. This is the exact transpose of the person-network pipeline: Pearson correlation was computed between every pair of items' 87-respondent answer vectors (rather than between respondents' 60-item vectors), and the resulting 60×60 correlation matrix was sparsified with the same mutual k-Nearest-Neighbours procedure ($k = 8$) used for the person network, for direct methodological comparability. Louvain community detection was then applied to the resulting item network, and the discovered communities were compared against the survey's own four a-priori topic labels using Adjusted Rand Index, to test whether real opinion structure follows the survey's topic design or cuts across it.

3 Analysis and Visualizations

3.1 Overall Network Structure

3.2 Topic Sub-Networks and Cross-Topic Comparison

Table 3 summarizes the four topic-specific sub-networks.

Metric	Value
Nodes	87
Edges	134
Density	0.036
Average clustering coefficient	0.157
Average path length (largest component)	3.86
Diameter (largest component)	9
Number of connected components	18
Largest component size	66
Number of Louvain communities	25
Modularity	0.578

Table 2: Overall opinion network metrics.

Topic	Edges	Density
Technology	201	0.054
Education	176	0.047
Ethics/Society	227	0.061
Environment	193	0.052

Table 3: Topic sub-network metrics (87 nodes each).

3.3 Opinion Bridges

To identify respondents who structurally connect otherwise separate opinion communities, betweenness centrality was computed for every node, and the community diversity of each top-scoring node's immediate neighborhood was examined (Table 4).

Respondent	Betweenness Centrality	Own Community	Neighbor Communities Spanned
81	0.083	16	5
35	0.090	12	3
90	0.062	12	4
55	0.074	18	3
121	0.083	0	2

Table 4: Top opinion-bridge respondents by betweenness centrality.

3.4 Community Opinion Archetypes

For every community with at least 4 members, the average deviation from the class-wide opinion score was computed per topic, producing interpretable "opinion archetypes" for each cluster (Figure 7).

3.5 Item-Opinion Network

Table 5 summarizes the item-opinion network constructed in Section 2.5.

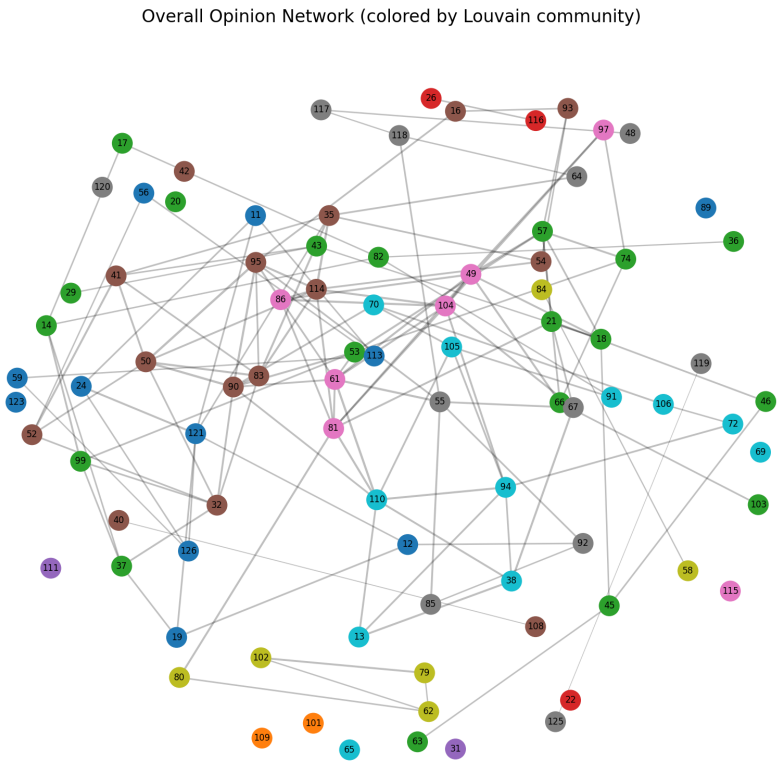


Figure 3: Overall opinion network, nodes colored by Louvain community.

[TODO: PRATHISH (Statistical Validation): insert null-model validation figures here (e.g. histogram of null-model modularity scores with observed modularity marked, significance test results).]

4 Results and Discussion

4.1 Person-Opinion Network (this workstream)

The overall opinion network fragments into approximately 25 Louvain communities at a modularity of 0.578, indicating that class opinion is organized into many small, fine-grained groups rather than 2–3 broad ideological camps.

Comparing topic-specific community structure to the overall network via Adjusted Rand Index shows that **Education** (ARI = 0.181) and **Technology** (ARI = 0.101) opinions are the strongest predictors of a respondent's overall opinion-community membership, while **Ethics/Society** (ARI = 0.023) and **Environment** (ARI = 0.030) opinions are essentially uncorrelated with overall clustering. This suggests that ethics and environmental questions attract broad, cross-cutting agreement or disagreement that does not track the same social groupings visible in tech/education opinions.

The bridge analysis further shows that the network's cohesion depends on a small number of specific individuals rather than broad consensus: respondent 81's immediate neighbors span 5 distinct communities, more than any other node, making this respondent (and the four others in

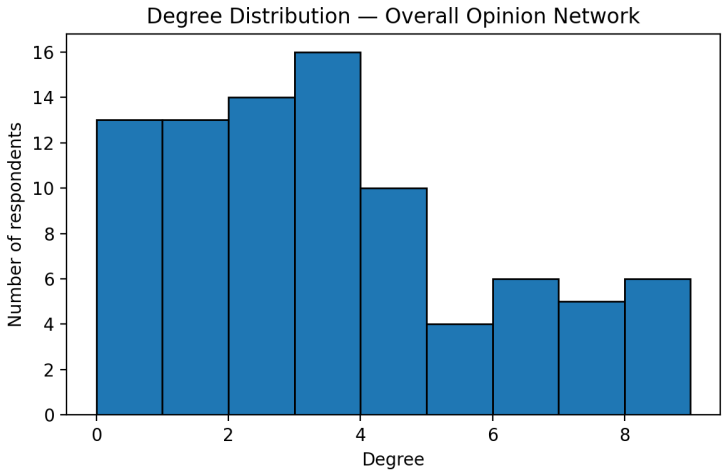


Figure 4: Degree distribution of the overall opinion network.

Metric	Value
Nodes (survey items)	60
Edges	140
Density	0.079
Average clustering coefficient	0.107
Average path length (largest component)	3.04
Diameter (largest component)	7
Number of connected components	4
Largest component size	57
Number of Louvain communities	9
Modularity	0.428
ARI vs. a-priori survey topics	0.143

Table 5: Item-opinion network metrics.

Table 4) structurally critical to keeping otherwise separate opinion groups connected.

The community opinion archetypes reveal clearly interpretable clusters: for example, community 0 (9 members) scores well below the class average on both Ethics/Society and Environment items – a coherent "skeptic" archetype – while community 16 (6 members) scores above average on every topic, representing a small "strong-consensus" cluster.

4.2 Item-Opinion Network (this workstream)

The item-opinion network is noticeably less cleanly separated than the person-opinion network: Louvain finds 9 communities at modularity ≈ 0.43 (versus 25 communities at modularity 0.578 for people), and three of those nine are singleton or near-singleton items with degree 0–1 – individual statements that nobody's answer pattern strongly tracks. Survey items, in other words, are more weakly and diffusely related to one another than respondents are to each other.

More importantly, the item communities do *not* reproduce the survey's own four-topic design: ARI between Louvain item communities and the a-priori topic labels is only 0.143. Instead of

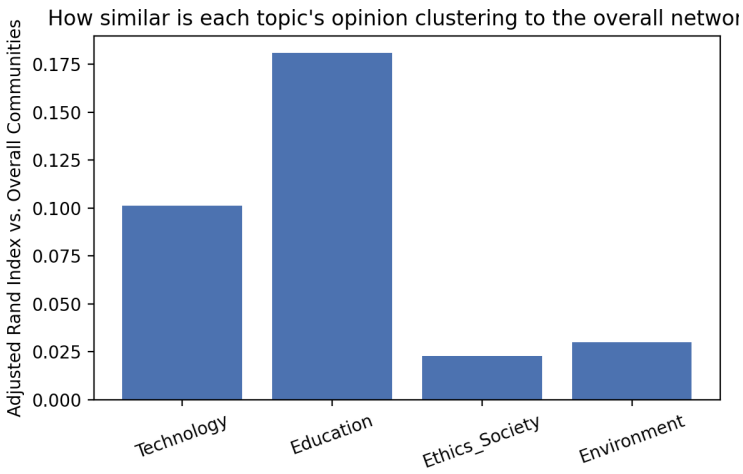


Figure 5: Adjusted Rand Index between each topic's community structure and the overall network's community structure.

Technology, Education, Ethics/Society, and Environment each forming their own tight cluster, the two largest communities each blend two nominal topics: one groups most Technology and Education items together (a "modernization / innovation-attitude" dimension – how someone feels about AI in the classroom correlates with how they feel about AI in society generally), while another groups most Ethics/Society and Environment items together (a "collective responsibility" dimension). This is a genuinely different axis of variation than the survey's own category labels would suggest.

This finding lines up closely with the person-network results above. Recall that Education and Technology opinions were the strongest predictors of a respondent's overall community membership (ARI 0.18 and 0.10), while Ethics/Society and Environment opinions were essentially uncorrelated with it (ARI $\approx$ 0.02–0.03). The item-level picture explains *why*: Ethics/Society and Environment items are exactly the ones with the highest agreement and lowest spread in the descriptive statistics (Table 1 – means of 1.28 and 1.39 out of 2, with the smallest standard deviations of any topic). When almost everyone agrees with an item, that item carries very little information about how respondents differ from one another, so it cannot drive person-level community structure even though it clusters tightly with other similarly-high-consensus items at the item level. Technology and Education items, by contrast, have more spread (std. 0.99 and 1.15) and therefore more genuine disagreement to organize people around – consistent with them being the topics that best predict overall person-community membership.

[TODO: PRATHISH (Statistical Validation): discuss null-model validation findings here – is the observed community structure statistically significant compared to random graphs of the same size/density? Does this justify $k = 8$ as a sparsification parameter?]

4.3 Overall Synthesis

[TODO: ALL: once each subsection above is written, add 1 short paragraph tying the three analyses together – what does the combination of the person-network, item-network, and statistical validation tell us about how this class thinks, collectively?]



Figure 6: Opinion bridges (red-ringed nodes) overlaid on the overall network.

5 Individual Contribution

- **Shreem Shukla**: Cleaned and pre-processed the raw survey data (missing-value handling, dropping incomplete responses, Likert-to-numeric encoding); built the full person-opinion network construction pipeline (Pearson similarity, mutual k-NN graph construction, Louvain community detection, degree/betweenness/ eigenvector centrality); performed the cross-topic ARI comparison, the opinion-bridge analysis, and the community opinion-archetype analysis. Wrote the Pipeline Followed section, the data-cleaning portion of Dataset Documentation, and this report skeleton.
- **Kris Jain**: Wrote up and elaborated the Dataset Documentation section (cleaning/encoding narrative, per-topic descriptive statistics, distribution plots, and a discussion of self-report survey bias). Designed and built the item-opinion network (`item_network/build_item_network.py`): item-item Pearson correlation, mutual k-NN graph construction, Louvain community detection, and an Adjusted Rand Index comparison against the survey's a-priori topic labels to test whether real opinion structure follows the survey's own topic design. Produced the item-network figures and metrics table, and wrote the corresponding Item-Opinion Network subsections of Pipeline Followed, Analysis and Visualizations, and Results and Discussion.
- **Prathish Ghosh**: **[TODO: fill in – e.g. "Designed and ran null-model statistical validation of network community structure; performed sensitivity analysis across k values; contributed to Results and Discussion."]**

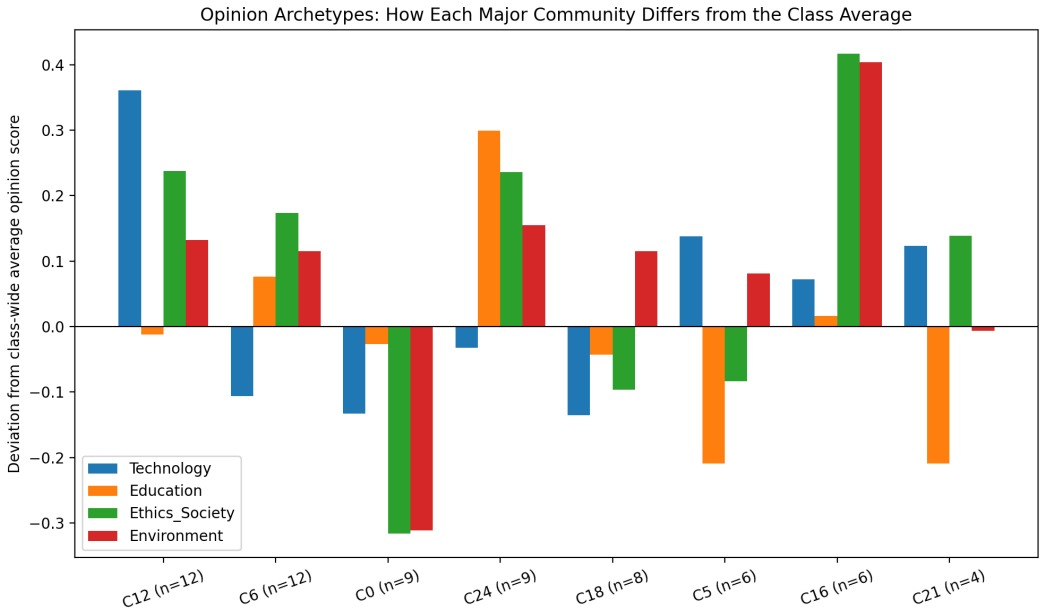


Figure 7: Opinion archetypes: deviation of each major community from the class-wide average, by topic.



Figure 8: Item-opinion network, nodes colored by Louvain community and labeled by item code (e.g. T01 = Technology item 1). Node layout is identical to Figure 9.

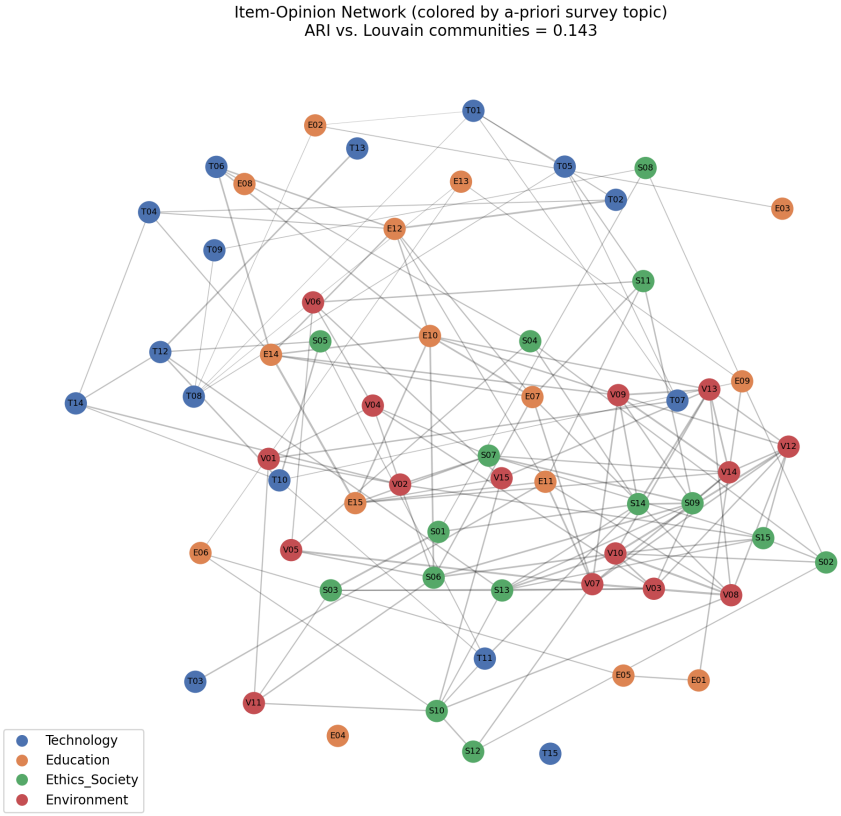


Figure 9: The same item-opinion network and layout as Figure 8, but colored by the survey's a-priori topic label instead of the discovered community. The visual mismatch between the two colorings reflects the low ARI (0.143) between them.

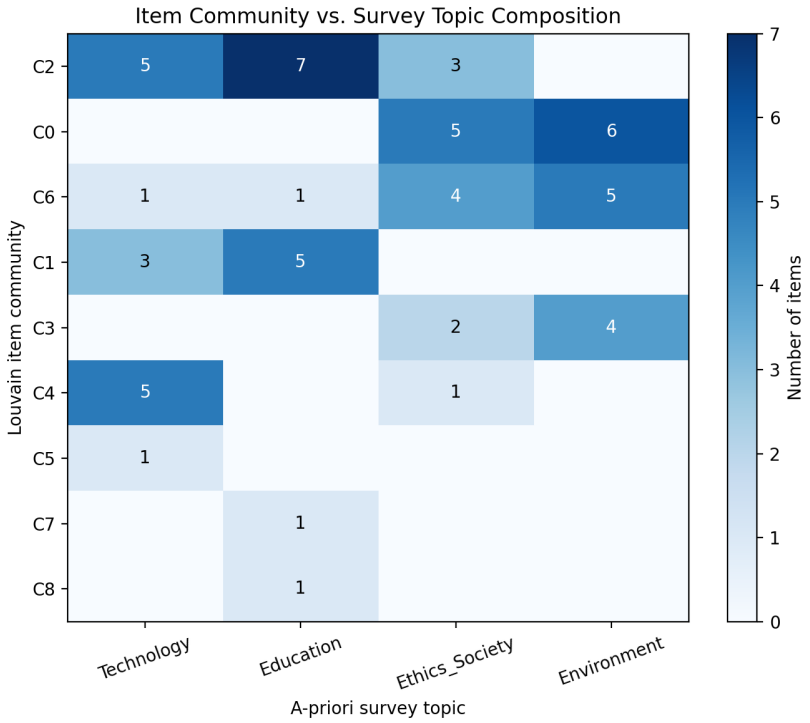


Figure 10: Number of items from each a-priori survey topic contained in each discovered Louvain item community. The two largest communities each draw items from two different topics rather than one.